



End Semester/Back Examination May -2025

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech. (Computer Science)
Semester/Year:	2
Paper Code:	BTEMACO204T22
Title of Paper:	Engineering Mathematics II.
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS/NOTES FOR THE EXAMINEE

1. This question paper consists of two sections, Section-A and Section-B.
 2. Section-A consists of seven questions and
 3. Section-B consists of six questions.
- Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question

Question
Number

SECTION - A

Marks
Allotted

Q.1

Define (i) Linear Dependent and linear independent vector.

2+2=4

(ii) Eigen values and Eigen vector

Q.2

Solve the Differential equation $x \frac{dy}{dx} + y - x^4 = 0$

4

Solve the following differential equation.

Q.3

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{y}{1+y^2}$$

4

Q.4

Obtain the Fourier expansion of $f(x) = x$, In the interval $(-\pi, \pi)$

4

Q.5

Find the diagonalization of matrix.

$$\begin{pmatrix} 1 & 2 & 3 \\ -6 & 7 & 8 \\ 2 & 4 & 6 \end{pmatrix}$$

4

Q.6

Obtain the partial differential equation by eliminating the arbitrary function. $z = f(x^2 + y^2)$

4

Q.7

Solve the following partial differential equation by Lagrange's method.

$$xzp + yzq = xy$$

SECTION - B

Q.1

Find the Fourier series of $f(x) = \sin px$, in the interval $-\pi \leq x \leq \pi$

10

Find the rank of the following matrix

Q.2

$$A = \begin{pmatrix} 1 & 2 & 3 \\ -6 & 7 & 8 \\ 3 & 4 & 5 \end{pmatrix}$$

10

Q.3

Solve the following differential equation.

$$(D^4 + 2D^3 - 3D^2)y = 4 \cos x$$

10

Solve the following Second order differential equation.

Q.4

$$\frac{d^2 y}{dx^2} - \cot x \frac{dy}{dx} - 2(1 - \cot x)y = e^x \sin x$$

10

Test the following system for consistency and solve

Q.5

$$\begin{aligned} x + y + z &= 6, & 2x + y + 3z &= 13, & 5x + 2y + z &= 12 \end{aligned}$$

10

Q.6

Solve the following partial differential equation by lagrange's method

$$(mz - ny)p + (nx - lz)q = ly - mx$$

10